91261 Practice Excellence Questions

Answer Schedule

Q	Answer
1	$x = 2$ or $x = -1$
2	$x = -1,\ 1,\ 2,\ 4$
3	$x = 6$
4	$\log_{ab} x = \frac{12}{7}$
5	$x = \frac{\log 30}{\log 1.5}$ (≈ 8.39)
6	After approximately 10.0 hours $(A \approx 1012,\ r \approx 1.581,\ t \approx 10.03)$
7	After 21 complete years
8	Approximately 13.5 m
9	$\Delta = k^2 + 2k + 9 = (k+1)^2 + 8 > 0$ for all k, so two distinct real roots
10	$\Delta = (a-b)^2 + 4k^2 \geq 0$, so always real solutions; exactly one solution when $a = b$ and $k = 0$
11	$m = 7$, touching at $(2, 13)$; $m = -1$, touching at $(-2, 1)$
12	$3 < m < 4$
13	$b = 7$ or $b = -7$
14	Shown: roots $2r$ and $3r$ give $5r = -\frac{b}{a}$ and $6r^2 = \frac{c}{a}$, so $6b^2 = 25ac$
15	$k = 0$ only
16	$k = 1$; not valid for $x = 4$ or $x = -4$
17	8 friends
18	$10\ \text{km h}^{-1}$
19	15 m
20	$x \approx 1.42$
21	$x = \frac{3}{2},\ y = \frac{9}{4}$
22	$x = 8,\ y = 4$
23	$x < -\sqrt{6}$ or $x > \sqrt{6}$
24	$x = 8$ or $x = 27$
25	$x = 0$ or $x = 1$
26	$x = 2$ or $x = 4$
27	Shown: $n(n+1)(n+2)(n+3) + 1 = \left(n^2 + 3n + 1\right)^2$
28	(i) 23 (ii) 110
29	$x = \frac{y+3}{y-2}$; y can never equal 2

Q	Answer
30	$k = 1$ or $k = -1$
31	$k = 0$ (solutions $x = -1,\ 0,\ 1$)
32	$10^{1.4} \approx 25.1$ times larger
33	Approximately 14 200 years ago ($\approx$ 14 176 years)
34	5 cm and 12 cm
35	$\frac{L}{W} = \frac{1+\sqrt{5}}{2}$
36	$\$10 \le p \le \$20$
37	Ella is not correct: reciprocal roots need $2k = 1$, so $k = \frac{1}{2}$, but then $\Delta = -\frac{15}{4} < 0$ (no real roots)
38	$x = 3$ or $x = -1$
39	$x = 100$ or $x = 0.1$
40	$k = 1$ or $k = 2$